

Lecture 1.7 – Matrices and Arrays

Specific Learning Objectives:

1.1.10 – Create vectors, arrays, matrices, lists, and data frames.

1.1.11 – Understand vectors and vectorized calculations.

1.1.12 – Learn how to index vectors, arrays, matrices, lists, and data frames.

Question

- What's the difference between a vector, a matrix, and an array?

- **Vectors** can be 1 or more elements in length, but must be 1-dimensional.

1 dimension


$$\vec{v} = \begin{bmatrix} 1 & 7 & 9 \end{bmatrix}$$

- **Matrices** are 2-dimensional.

1 dimension



1 dimension


$$\mathbf{m} = \begin{bmatrix} 1 & 7 & 9 \\ 2 & 6 & 3 \\ 8 & 1 & 4 \end{bmatrix}$$

- **Arrays** are n -dimensional.

Example: 3-D array

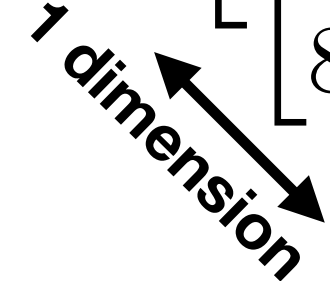
1 dimension



1 dimension



1 dimension


$$\mathbf{a} = \begin{bmatrix} \begin{bmatrix} 2 & 4 & 7 \\ 8 & 1 & 9 \end{bmatrix} & \begin{bmatrix} 7 & 1 & 9 \\ 1 & 0 & 3 \end{bmatrix} \\ \begin{bmatrix} 4 & 2 & 7 \\ 9 & 1 & 2 \end{bmatrix} & \begin{bmatrix} 6 & 2 & 8 \end{bmatrix} \end{bmatrix}$$

Matrix data

- **Matrices** are just several vectors stored together!

$$\vec{v} = \begin{bmatrix} 1 & 7 & 9 \end{bmatrix} \quad m = \begin{bmatrix} \begin{bmatrix} 1 & 7 & 9 \end{bmatrix} \\ \begin{bmatrix} 2 & 6 & 3 \end{bmatrix} \\ \begin{bmatrix} 8 & 1 & 4 \end{bmatrix} \end{bmatrix} \begin{matrix} \text{vector 1 (v1)} \\ \text{vector 2 (v2)} \\ \text{vector 3 (v3)} \end{matrix}$$

- Create a matrix in R with `matrix()`:

```
> v1 <- c(1,7,9)
> v2 <- c(2,6,3)
> v3 <- c(8,1,4)
> m <- matrix(c(v1,v2,v3), nrow=3, byrow=TRUE)
> m
```

	[,1]	[,2]	[,3]	
[1,]	1	7	9	= v1
[2,]	2	6	3	= v2
[3,]	8	1	4	= v3

Creating Matrices

- Creating matrices with `matrix()` provides many options!

```
> v1 <- c(1,7,9)
> v2 <- c(2,6,3)
> v3 <- c(8,1,4)
```

Number of rows the
matrix should be

```
> m1 <- matrix(c(v1,v2,v3), nrow=3, byrow=TRUE)
> m1
```

	[,1]	[,2]	[,3]
[1,]	1	7	9
[2,]	2	6	3
[3,]	8	1	4

Construct the matrix
by row (TRUE) or by
column (FALSE)

```
> m2 <- matrix(c(v1,v2,v3), nrow=3, byrow=FALSE)
> m2
```

	[,1]	[,2]	[,3]
[1,]	1	2	8
[2,]	7	6	1
[3,]	9	3	4

**By column is
default behavior**

Matrix dimensions

- Several functions help with determining the size of matrices.

```
> big.m <- cbind(m1,m2)
> big.m
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	1	7	9	1	2	8
[2,]	2	6	3	7	6	1
[3,]	8	1	4	9	3	4

- **length()** will give you the total number of elements

```
> length(big.m)
[1] 18
```

- **dim()** will give you the dimensions of the matrix:

```
> dim(big.m)           > dim(big.m)[2]
[1] 3 6                [1] 6
```

- **nrow()** will give you the number of rows:

```
> nrow(big.m)
[1] 3
```

- **ncol()** will give you the number of columns:

```
> ncol(big.m)
[1] 6
```

Check Your Understanding

You have four vectors in your environment:

```
v1 <- c(1, 1, 1)
```

```
v3 <- c(3, 3, 3)
```

```
v2 <- rep(2, 3)
```

```
v4 <- rep(4, 3)
```

You run the following line to create a matrix:

```
my.matrix <- matrix(c(v1,v2,v3,v4), nrow=4, byrow=FALSE)
```

Which describes the output matrix `my.matrix`?

```
> matrix(c(v1,v2,v3,v4),nrow=3,byrow=FALSE)
```

a)

	[,1]	[,2]	[,3]	[,4]
[1,]	1	2	3	4
[2,]	1	2	3	4
[3,]	1	2	3	4

```
> matrix(c(v1,v2,v3,v4),nrow=3,byrow=TRUE)
```

c)

	[,1]	[,2]	[,3]	[,4]
[1,]	1	1	1	2
[2,]	2	2	3	3
[3,]	3	4	4	4

Correct

b)

	[,1]	[,2]	[,3]
[1,]	1	2	3
[2,]	1	2	4
[3,]	1	3	4
[4,]	2	3	4

```
> matrix(c(v1,v2,v3,v4),nrow=4,byrow=TRUE)
```

d)

	[,1]	[,2]	[,3]
[1,]	1	1	1
[2,]	2	2	2
[3,]	3	3	3
[4,]	4	4	4

Indexing a Matrix

- **Indexing a matrix:** use the position of both row and column to pick out an element.

row elements

m[• , •]

column elements

(Leave column spot empty to grab everything in that column)

> m

	[,1]	[,2]	[,3]
[1,]	1	7	9
[2,]	2	6	3
[3,]	8	1	4

Indexing by row: m[1,]

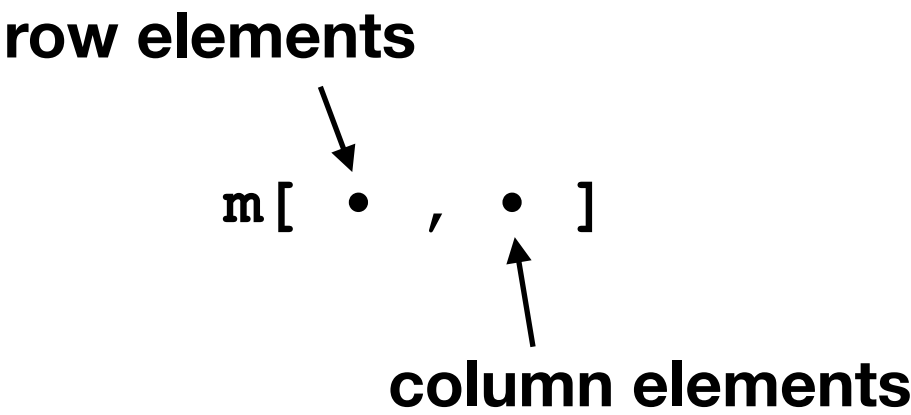
m[1,] = v1

m[2,] = v2

m[3,] = v3

Indexing a Matrix

- **Indexing a matrix:** use the position of both row and column to pick out an element.



Indexing by column: m[, 1]

What are m[, 1], m[, 2], and m[, 3]?

> m

	[, 1]	[, 2]	[, 3]
[1,]	1	7	9
[2,]	2	6	3
[3,]	8	1	4

Indexing a single element:

m[2 , 3] =
3

Subsetting Matrices

- Similar to vectors, you can subset info in a matrix by specifying which rows and/or columns you want.

```
> big.m <- cbind(m1,m2)
```

```
> big.m
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	1	7	9	1	2	8
[2,]	2	6	3	7	6	1
[3,]	8	1	4	9	3	4

- Subset a number of columns or rows with positive integers:

```
> big.m[,c(1,3,4)]
```

	[,1]	[,2]	[,3]
[1,]	1	9	1
[2,]	2	3	7
[3,]	8	4	9

Columns
1, 3, & 4

```
> big.m[,1:4]
```

	[,1]	[,2]	[,3]	[,4]
[1,]	1	7	9	1
[2,]	2	6	3	7
[3,]	8	1	4	9

Columns 1
through 4

- Remove a number of columns or rows with negative integers:

```
> big.m[-1,]
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	2	6	3	7	6	1
[2,]	8	1	4	9	3	4

All rows
except 1

Check Your Understanding

You run the following line to create a matrix:

```
my.matrix2 <- matrix(seq(1,21), nrow=7, byrow=TRUE)
```

Which line of code will subset only two-digit numbers (those greater than 9)?

Correct

a) `my.matrix2[4:7, ]`

```
> my.matrix2[4:7,]  
      [,1] [,2] [,3]  
[1,]   10   11   12  
[2,]   13   14   15  
[3,]   16   17   18  
[4,]   19   20   21
```

b) `my.matrix2[4:7, 3]`

```
> my.matrix2[4:7,3]  
[1] 12 15 18 21
```

c) `my.matrix2[>9]`

```
> my.matrix2[>9]  
Error: unexpected '>' in "my.matrix2[>"
```

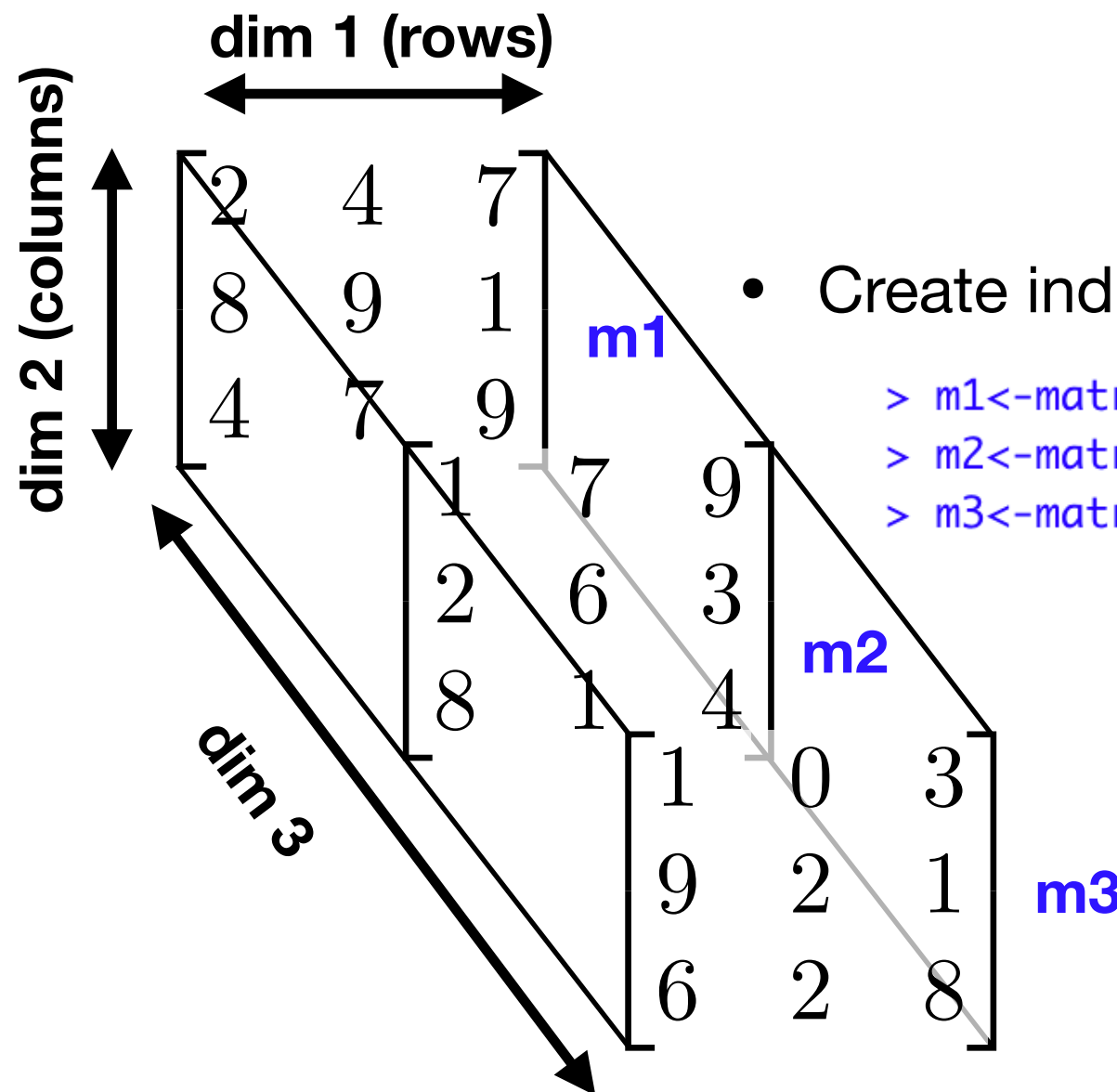
d) `my.matrix2[, 4:7]`

```
> my.matrix2[,4:7]  
Error in my.matrix2[, 4:7] :  
subscript out of bounds
```

Multidimensional Arrays

- **Arrays** are n -dimensional and numeric.

Example: 3-D array

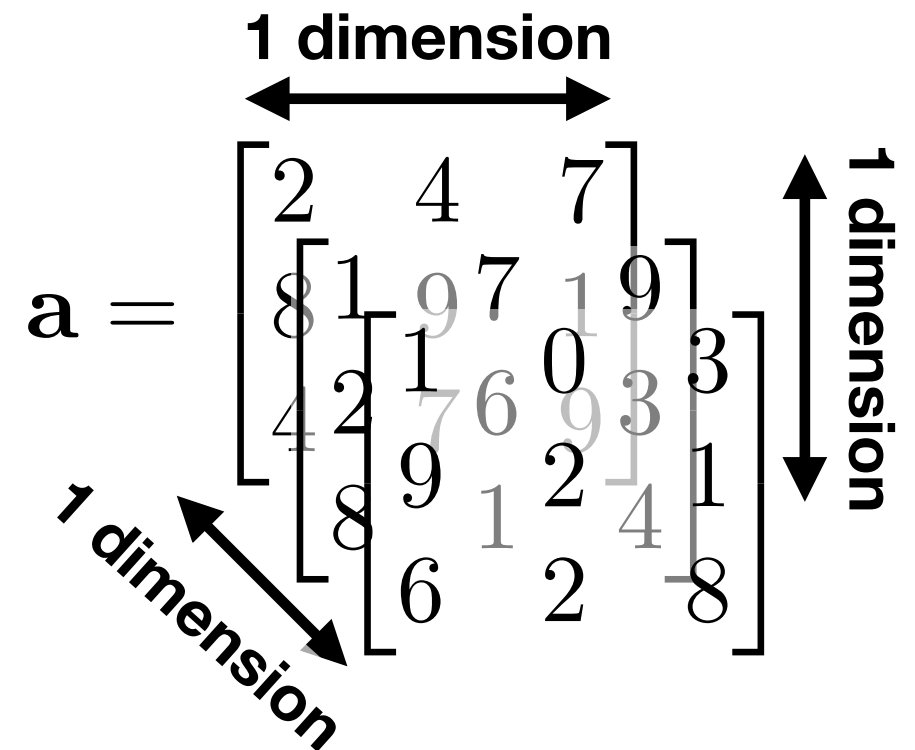


- Create individual matrices:

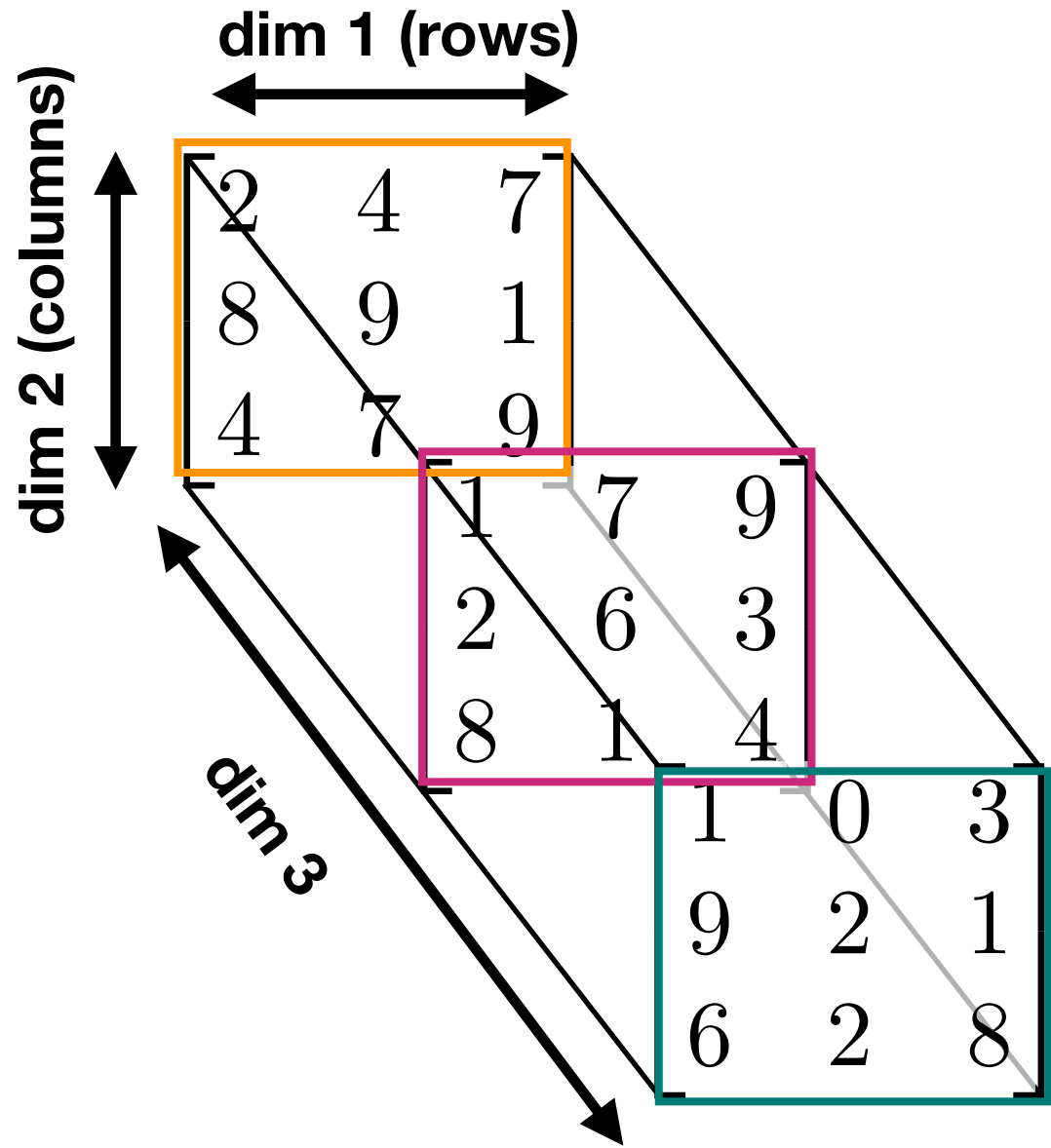
```
> m1<-matrix(c(2,4,7,8,9,1,4,7,9), nrow=3, byrow=TRUE)
> m2<-matrix(c(1,7,9,2,6,3,8,1,4), nrow=3, byrow=TRUE)
> m3<-matrix(c(1,0,3,9,2,1,6,2,8), nrow=3, byrow=TRUE)
```

- Create a 3D array:

```
> a1 <- array(c(m1,m2,m3),dim=c(3,3,3))
```



Indexing Multidimensional Arrays



> a1
, , 1

	[,1]	[,2]	[,3]
[1,]	2	4	7
[2,]	8	9	1
[3,]	4	7	9

, , 2

	[,1]	[,2]	[,3]
[1,]	1	7	9
[2,]	2	6	3
[3,]	8	1	4

, , 3

	[,1]	[,2]	[,3]
[1,]	1	0	3
[2,]	9	2	1
[3,]	6	2	8

Check Your Understanding

```
> a1  
, , 1
```

	[,1]	[,2]	[,3]
[1,]	2	4	7
[2,]	8	9	1
[3,]	4	7	9

```
, , 2
```

	[,1]	[,2]	[,3]
[1,]	1	7	9
[2,]	2	6	3
[3,]	8	1	4

```
, , 3 column
```

row →

	[,1]	[,2]	[,3]
[1,]	1	0	3
[2,]	9	2	1
[3,]	6	2	8

Which position in a1 holds the value 0?

a) a1[1, 1, 1] c) a1[2, 1, 3]

b) a1[1, 2, 3] d) a1[1, 2, 1]

Correct

Which positions in a1 hold the value 7?

Write down your answer!

Action Items

- 1. Complete assignments 1.8 and 1.9.**
- 2. Read Davies Ch. 5 for next time.**